

SQL ANALYSIS APPENDIX

Telco Customer Churn Analysis

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This appendix contains all SQL queries written and executed in Google BigQuery as part of the Telco Customer Churn Analysis case study. Each query includes its analytical intent, full code, results, and key finding. Queries progress from a broad overview of churn to increasingly specific risk profiling.

QUERY 1 — Overall Churn Rate

? **Intent:** To establish the baseline business problem by calculating what percentage of the total customer base has churned versus remained active. This number becomes the headline metric for the entire analysis.

```
SELECT
  `Churn Label`,
  COUNT(*) AS total_customers,
  ROUND(COUNT(*) * 100.0 / SUM(COUNT(*) OVER()), 2) AS percentage
FROM `our-signal-496317-q7.telco_churn.customer_data`
GROUP BY `Churn Label`
```

RESULTS

Churn Label	total_customers	percentage
false	5,174	73.46%
true	1,869	26.54%

! **Key Finding:** 26.54% of customers have churned — roughly 1 in 4. This sits at the high end of the telecom industry average of 20-25% and establishes the urgency of the retention problem.

QUERY 2 — Churn by Contract Type

? **Intent:** To determine whether the type of contract a customer holds influences their likelihood of churning. This tests the hypothesis that customers with less commitment (month-to-month) are more likely to leave than those locked into longer agreements.

```
SELECT
  Contract,
  COUNT(*) AS total_customers,
  SUM(CASE WHEN `Churn Label` = true THEN 1 ELSE 0 END) AS churned_customers,
  ROUND(SUM(CASE WHEN `Churn Label` = true THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*), 2) AS churn_rate
FROM `our-signal-496317-q7.telco_churn.customer_data`
GROUP BY Contract
ORDER BY churn_rate DESC
```

RESULTS

Contract	total_customers	churned_customers	churn_rate
Month-to-month	3,875	1,655	42.71%
One year	1,473	166	11.27%
Two year	1,695	48	2.83%

! **Key Finding:** Contract type is the strongest single predictor of churn. Month-to-month customers churn at 42.71% — 15x higher than two-year contract holders at 2.83%. Converting customers to longer contracts is the single highest-leverage retention action available.

QUERY 3 — Churn by Tenure Group

? **Intent:** To understand how long a customer has been with the company affects their likelihood of leaving. Customers were grouped into four lifecycle stages using CASE WHEN binning to reveal whether newer or longer-tenured customers are at higher risk.

```
SELECT
  CASE
    WHEN `Tenure Months` <= 12 THEN '0-12 Months (New)'
    WHEN `Tenure Months` <= 24 THEN '13-24 Months (Developing)'
    WHEN `Tenure Months` <= 48 THEN '25-48 Months (Established)'
    ELSE '49+ Months (Loyal)'
  END AS tenure_group,
  COUNT(*) AS total_customers,
  SUM(CASE WHEN `Churn Label` = true THEN 1 ELSE 0 END) AS churned_customers,
```

```
ROUND(SUM(CASE WHEN `Churn Label` = true THEN 1 ELSE 0 END)
  * 100.0 / COUNT(*), 2) AS churn_rate
FROM `our-signal-496317-q7.telco_churn.customer_data`
GROUP BY tenure_group
ORDER BY churn_rate DESC
```

RESULTS

Tenure Group	total_customers	churned_customers	churn_rate
0-12 Months (New)	2,186	1,037	47.44%
13-24 Months (Developing)	1,024	294	28.71%
25-48 Months (Established)	1,594	325	20.39%
49+ Months (Loyal)	2,239	213	9.51%

! Key Finding: The first 12 months are the highest-risk window — nearly half of new customers (47.44%) leave before completing their first year. Churn drops dramatically as tenure increases, falling to just 9.51% for loyal customers of 49+ months. Early onboarding and retention programs targeting the first year are critical.

QUERY 4 — Churn by Monthly Charges

? Intent: To investigate whether pricing level drives customer departure. Customers were segmented into four pricing tiers using CASE WHEN binning to identify whether higher charges correlate with higher churn rates.

```
SELECT
  CASE
    WHEN `Monthly Charges` <= 35 THEN 'Low ($0-$35)'
    WHEN `Monthly Charges` <= 65 THEN 'Medium ($36-$65)'
    WHEN `Monthly Charges` <= 85 THEN 'High ($66-$85)'
    ELSE 'Very High ($86+)'
  END AS charge_group,
  COUNT(*) AS total_customers,
  SUM(CASE WHEN `Churn Label` = true THEN 1 ELSE 0 END) AS churned_customers,
  ROUND(SUM(CASE WHEN `Churn Label` = true THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*), 2) AS churn_rate
FROM `our-signal-496317-q7.telco_churn.customer_data`
GROUP BY charge_group
ORDER BY churn_rate DESC
```

RESULTS

Charge Group	total_customers	churned_customers	churn_rate
High (\$66-\$85)	1,711	614	35.89%
Very High (\$86+)	2,188	740	33.82%
Medium (\$36-\$65)	1,409	326	23.14%
Low (\$0-\$35)	1,735	189	10.89%

! Key Finding: The \$66-\$85 tier has the highest churn rate (35.89%), slightly above the \$86+ tier (33.82%). This suggests customers in the high tier feel overcharged without receiving premium value, while customers paying the most may be on bundled packages that justify the cost. The \$66-\$85 pricing tier has a perceived price-value mismatch that needs addressing.

QUERY 5 — Top 10 Churn Reasons

? Intent: To understand the 'why' behind customer departures by analyzing the Churn Reason column. A WHERE filter was applied to isolate only churned customers with a recorded reason, and results were limited to the top 10 most frequent responses to identify dominant themes.

```
SELECT
  `Churn Reason`,
  COUNT(*) AS total_customers,
  ROUND(COUNT(*) * 100.0 / SUM(COUNT(*) OVER(), 2) AS percentage
FROM `our-signal-496317-q7.telco_churn.customer_data`
WHERE `Churn Label` = true
  AND `Churn Reason` IS NOT NULL
GROUP BY `Churn Reason`
ORDER BY total_customers DESC
LIMIT 10
```

RESULTS

Churn Reason	customers	percentage
Attitude of support person	192	10.27%
Competitor offered higher download speeds	189	10.11%
Competitor offered more data	162	8.67%
Don't know	154	8.24%
Competitor made better offer	140	7.49%
Attitude of service provider	135	7.22%
Competitor had better devices	130	6.96%

Network reliability	103	5.51%
Product dissatisfaction	102	5.46%
Price too high	98	5.24%

! Key Finding: Three themes dominate churn: (1) Competition accounts for ~43% combined — customers are being poached by competitors with better speeds, data, offers, and devices. (2) Poor customer service attitude is the single largest individual reason at 10.27% — entirely within the company's control. (3) Product and price issues account for the remaining top reasons. The company has both a service culture problem and a competitive positioning problem.

QUERY 6 — High Risk Customer Profile

? Intent: To build a definitive high-risk customer profile by combining three variables simultaneously — contract type, tenure, and monthly charges. This multi-dimensional analysis identifies which exact combination of factors creates the highest churn risk, giving the business a precise target for intervention.

```
SELECT
  Contract,
  CASE
    WHEN `Tenure Months` <= 12 THEN '0-12 Months (New)'
    WHEN `Tenure Months` <= 24 THEN '13-24 Months (Developing)'
    WHEN `Tenure Months` <= 48 THEN '25-48 Months (Established)'
    ELSE '49+ Months (Loyal)'
  END AS tenure_group,
  CASE
    WHEN `Monthly Charges` <= 35 THEN 'Low ($0-$35)'
    WHEN `Monthly Charges` <= 65 THEN 'Medium ($36-$65)'
    WHEN `Monthly Charges` <= 85 THEN 'High ($66-$85)'
    ELSE 'Very High ($86+)'
  END AS charge_group,
  COUNT(*) AS total_customers,
  SUM(CASE WHEN `Churn Label` = true THEN 1 ELSE 0 END) AS churned_customers,
  ROUND(SUM(CASE WHEN `Churn Label` = true THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*), 2) AS churn_rate
FROM `our-signal-496317-q7.telco_churn.customer_data`
GROUP BY Contract, tenure_group, charge_group
ORDER BY churn_rate DESC
LIMIT 10
```

RESULTS

Contract	Tenure Group	Charge Group	customers	churn_rate
Month-to-month	0-12 Months	Very High (\$86+)	327	74.92%

Month-to-month	0-12 Months	High (\$66-\$85)	648	64.04%
Month-to-month	13-24 Months	Very High (\$86+)	251	55.78%
Month-to-month	25-48 Months	Very High (\$86+)	364	42.86%
Month-to-month	0-12 Months	Medium (\$36-\$65)	510	42.55%
Month-to-month	13-24 Months	High (\$66-\$85)	216	36.11%
Month-to-month	25-48 Months	High (\$66-\$85)	207	35.75%
Month-to-month	49+ Months	Very High (\$86+)	218	30.28%
Month-to-month	13-24 Months	Medium (\$36-\$65)	159	28.93%
Month-to-month	0-12 Months	Low (\$0-\$35)	509	28.88%



Key Finding: Every single entry in the top 10 highest risk combinations is a month-to-month customer — confirming contract type as the dominant churn predictor regardless of tenure or price. The definitive high-risk profile is: a new month-to-month customer (0-12 months) paying over \$66/month, with a churn rate between 64-75%. This customer can be identified at sign-up and targeted immediately for retention intervention.